UNIVERSITY OF PERPETUAL HELP SYSTEM DALTA
COLLEGE OF COMPUTER STUDIES

ASCENOVA: AN INTELLIGENT GAMIFIED FITNESS TRACKING AND PERSONALIZED EXERCISE RECOMMENDATION SYSTEM

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Corpuz, Carl Justin
Ligtas, Aaron Jeremiah
Padilla, Christan

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Chapter 1

PROBLEMS AND ITS BACKGROUND

INTRODUCTION

Physical inactivity and the difficulty of maintaining consistent exercise routines have become common concerns in today's technology-driven environment. Many individuals have limited opportunities to engage in regular physical activity due to academic responsibilities, work-related tasks, prolonged use of digital devices, and other daily activities. Existing fitness applications provide users with tools for monitoring physical activities and managing their fitness routines; however, some systems may provide limited personalization and engagement based on the individual needs and goals of users.

These concerns highlight the need for a fitness system that can provide more personalized and engaging support for users. An intelligent fitness system can utilize user information and activity patterns to provide exercise recommendations suited to individual needs, while gamification features can encourage participation and continued engagement. In response to these concerns, the researchers propose ASCENOVA: An Intelligent Gamified Fitness Tracking and Personalized Exercise Recommendation System, which integrates personalized exercise recommendations, fitness

tracking, and gamification features to provide a more adaptive and engaging fitness management experience.

The rise of inactive lifestyles has become a significant concern for health and wellness in today's technology-driven world. Many people now spend long hours on digital devices, handling academic work, and completing job-related tasks, which limits their chances for regular physical activity. The World Health Organization (WHO, 2022) notes that physical inactivity is still one of the top risk factors linked to noncommunicable diseases like cardiovascular diseases, diabetes, and other health issues. Even though most people know about the benefits of exercise, many still struggle to build and stick to consistent fitness routines. This is often due to limited knowledge, low motivation, and uncertainty about which exercises fit their personal goals and physical conditions.

Existing fitness programs and mobile apps usually offer standard workout routines that do not fully meet the specific needs of each user. The effectiveness of exercise depends on several factors, such as age, body composition, physical ability, lifestyle, and desired fitness results. However, many fitness applications mainly focus on tracking activities like steps, calories, and workout history, instead of giving tailored recommendations. Because of this, users might up

with less effective exercise habits, slow progress, and lower motivation when they do not see meaningful results.

Recent advancements in artificial intelligence (AI) and data-driven technologies have created new opportunities for developing smarter and more adaptive fitness solutions. AI-powered systems are capable of analyzing user information, identifying activity patterns, and generating personalized exercise recommendations based on individual fitness profiles and goals. These intelligent capabilities allow fitness applications to provide more relevant exercise guidance, helping users make informed decisions and maintain healthier lifestyles. According to Lopez-Barreiro et al. (2024), AI-powered recommender systems play an important role in promoting healthy habits by delivering personalized recommendations that encourage regular physical activity and improve long-term user engagement.

Even though there are many fitness tracking applications available today, most existing systems still focus on monitoring users' physical activities while providing limited engagement and motivation. As a result, many users struggle to maintain consistent exercise routines because repetitive activities and minimal interaction reduce their interest over time. To address this challenge, gamification has become an effective strategy for improving user engagement in health and fitness applications. By incorporating

various gamification elements into health and fitness applications, developers can create a more interactive and enjoyable user experience that encourages users to participate consistently in physical activities. According to Yin et al. (2022), gamification elements have a positive influence on user satisfaction and contribute to a more engaging experience in health and fitness applications.

Therefore, this study focuses on the development of **ASCENOVA: An Intelligent Gamified Fitness Tracking and Personalized Exercise Recommendation System**. The proposed system aims to address the limitations of existing fitness applications by integrating artificial intelligence for personalized exercise recommendations, fitness activity tracking, and gamification features into a mobile application for end-users and a web application that provides the end-user features available on the mobile application, together with an admin dashboard for authorized administrators. By combining intelligent recommendations with interactive motivational elements, ASCENOVA seeks to provide users with a more personalized, engaging, and effective approach to fitness management while encouraging the adoption of healthier lifestyle habits.

Statement of the Problem

The rise of digital fitness applications has given people convenient ways to monitor their physical activities and manage their health goals. Still, many current fitness tracking systems mainly concentrate on logging workout data like steps, calories, and exercise history, but they offer only limited personalization and smart guidance. Users often find it challenging to choose suitable exercises, stay motivated, and keep up long-term fitness routines because these systems lack adaptive recommendations and engaging features.

This study intends to tackle these issues by developing ASCENOVA: An Intelligent Gamified Fitness Tracking and Personalized Exercise Recommendation System. This system brings together intelligent exercise suggestions, fitness activity tracking, and gamification features to create a more tailored and engaging fitness experience.

To address the identified gap, the researchers recommend that ASCENOVA generate exercise recommendations that adapt when the plan is regenerated using the user's recent activity, rather than relying solely on the fitness information initially provided during profile setup. When the user requests a new plan, the system considers the user's fitness profile together with a summary of the user's activity over the previous 14 days, including streak and weight

trend, so that the recommendations can remain relevant as the user's fitness level and habits change over time, thereby improving the personalization and long-term usefulness of the exercise guidance provided by the system.

1. How can ASCENOVA be designed based on the identified functional and non-functional requirements for the mobile application and web application?
2. How can ASCENOVA be developed to provide personalized exercise recommendations, fitness tracking, gamification features, and administrative functions through its mobile application and web application?
3. How can the developed ASCENOVA system be evaluated in terms of software quality based on ISO/IEC 9126, specifically Functionality, Efficiency, Usability, Reliability, Maintainability, and Portability, as well as its perceived effectiveness in providing personalized exercise recommendations, supporting fitness tracking, and promoting user engagement?

Objectives of the Study

General Objective

The general objective of this study is to design, develop, and evaluate ASCENOVA: An Intelligent Gamified Fitness Tracking and Personalized Exercise Recommendation System.

Specific Objectives

Specifically, this study aims to:

1. Design ASCENOVA based on the identified functional and non-functional requirements to provide personalized exercise recommendations, fitness tracking, and gamification features through a mobile application, and through a web application that provides the end-user features available on the mobile application, together with an admin dashboard for authorized administrators.
2. Develop ASCENOVA to provide personalized exercise recommendations based on users' fitness information, fitness goals, and recent activity patterns, adapting when the plan is regenerated, through the mobile application and the web application, while providing administrative functions through the admin dashboard for managing user accounts and achievements and for monitoring user activity, system usage, and AI usage.
3. Evaluate the developed ASCENOVA system in terms of its software quality based on ISO/IEC 9126, specifically Functionality, Efficiency,

Usability, Reliability, Maintainability, and Portability, as well as its perceived effectiveness in providing personalized exercise recommendations, supporting fitness tracking, and promoting user engagement.

Functional and Non-Functional Requirements

The functional and non-functional requirements of ASCENOVA were identified based on the objectives of the study and serve as the basis for the design and development of the mobile application and web application.

Functional Requirements

1. The system shall allow users to register, log in, and manage their fitness profile, including their fitness information, goals, and exercise preferences.
2. The system shall generate personalized exercise recommendations through an external artificial intelligence service/API based on the user's fitness information, goals, preferences, and activity patterns.
3. The system shall allow users to record their workout activities and monitor their fitness progress over time.
4. The system shall provide gamification features, including points, achievements, challenges, streaks, and leaderboards, to encourage user engagement.

5. The system shall allow authorized administrators to manage user accounts and achievements through the admin dashboard of the web application.
6. The system shall allow administrators to monitor user activity, system usage, and AI usage through the admin dashboard of the web application.
7. The system shall provide an AI Coach chat feature powered by an external artificial intelligence service/API.
8. The system shall provide a fallback library of pre-written workout plan templates, matched to the user's goal, fitness level, equipment, and preferred training days, which is used when the external artificial intelligence service fails to generate a workout plan.

Non-Functional Requirements

1. Efficiency The system shall respond to user actions and generate exercise recommendations within an acceptable amount of time while using system resources efficiently.
2. Usability – The system shall provide clear, understandable, and easy-to-navigate interfaces that support effective interaction for end-users and administrators across the mobile and web applications.
3. Reliability – The system shall perform its intended functions consistently and correctly, including providing workout-plan templates through the fallback library when the external AI service fails to generate a workout plan.
4. Functionality, Security – The system shall protect user accounts and personal fitness information through appropriate authentication, authorization, and access-control mechanisms.
5. Functionality – The system shall provide the functions necessary for personalized exercise recommendations, fitness tracking, gamification, and administrative management as identified in the study.
6. Portability - The system shall support its intended use across the ASCENOVA mobile application for end-users and the web application for end-users and administrators, with functions and interfaces appropriate to each platform.

7. Maintainablity - The system shall be structured to facilitate the identification and correction of errors, modification of existing functions, and implementation of future improvements without unnecessarily affecting other system components.

SCOPE AND LIMITATIONS

Scope of the Study

This study focuses on the design, development, and evaluation of ASCENOVA: An Intelligent Gamified Fitness Tracking and Personalized Exercise Recommendation System. The system consists of two main components: a mobile application for end-users and a web application that provides the end-user features available on the mobile application, together with an admin dashboard for authorized administrators.

The ASCENOVA mobile application is designed to allow users to create and manage their fitness profiles, provide their fitness information and goals, receive personalized exercise recommendations through an external artificial intelligence API, record their workout activities, monitor their fitness progress, and access gamification features such as points, achievements, challenges, streaks, leaderboards, and progress tracking.

The ASCENOVA web application provides the end-user features available on the mobile application, together with an admin

dashboard for authorized administrators. The admin dashboard allows administrators to manage user accounts and achievements and to monitor user activity, system usage, and AI usage.

ASCENOVA also includes an AI Coach chat feature powered by the external artificial intelligence API. As a reliability feature, the system includes a fallback library of 60 pre-written workout plan templates, matched to the user's goal, fitness level, equipment, and preferred training days, which is used when the external artificial intelligence API fails to generate a workout plan.

The developed ASCENOVA system will be evaluated using six selected product quality characteristics based on ISO/IEC 9126-1:2001: Functionality, Efficiency, Usability, Reliability, Maintainability, and Portability, with Security evaluated as a sub-characteristic of Functionality. The study will also assess the perceived effectiveness of the system in providing personalized exercise recommendations, supporting fitness tracking, and promoting user engagement.

Limitations of the Study

The study is limited to the development of the ASCENOVA mobile application for end-users and the ASCENOVA web application, which provides the end-user features together with an admin dashboard for administrative functions. The system is intended to provide general fitness guidance and should not be

considered a substitute for professional medical advice, diagnosis, or treatment.

The personalized exercise recommendations generated by the system depend on the accuracy and completeness of the information provided by users. Inaccurate or incomplete user information may affect the relevance of the recommended exercises.

The system does not include personalized meal planning, wearable device integration, real-time health monitoring, or rehabilitation programs. These features may be considered for future development. Nutrition support is limited to daily calorie, macronutrient, and water intake targets, and the system does not provide meal planning. The exercise recommendations will also be reviewed or validated with the assistance of a qualified fitness expert to assess their appropriateness for the intended users.

SIGNIFICANCE OF THE STUDY

This study is expected to benefit the following:

Fitness Enthusiasts

The system can help fitness enthusiasts by providing personalized exercise recommendations, tracking their workout progress, and motivating them through gamification features.

Students

Students can use the system as a guide in managing their physical activities despite their busy academic schedules. It may also encourage them to develop healthier exercise habits.

Working Individuals

The system may help working individuals maintain an active lifestyle by providing exercise recommendations that match their fitness goals and allowing them to monitor their progress.

Researchers

The findings of this study may serve as a reference for future researchers who plan to conduct similar studies related to artificial intelligence, fitness tracking systems, gamification, and personalized recommendation systems.

Future Developers

The developed system may serve as a basis for improving future fitness applications by adding more advanced features and technologies.

System administrators

System administrators may benefit from the admin dashboard of the web application by having a centralized platform for managing user accounts and achievements and for monitoring user activity, system usage, and AI usage. The administrative functions may help improve the organization and monitoring of information within the ASCENOVA system.

Fitness Experts

Fitness experts may benefit from the system by serving as validators of the exercise recommendations generated by ASCENOVA. Their expertise may help assess the appropriateness and suitability of the recommended exercises for the intended users.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the review of related literature and studies gathered from both local and international sources relevant to the present study. It discusses existing literature and previous studies related to artificial intelligence, fitness tracking, personalized exercise recommendation, and gamification. This chapter also presents the conceptual framework and the operational and technical definition of terms used in the study.

Foreign Literature

Lopez-Barreiro et al. (2024) conducted a systematic review on artificial intelligence-powered recommender systems for promoting healthy habits and active aging. The researchers reviewed different AI-based recommendation systems used in health and wellness applications and found that artificial intelligence can provide personalized recommendations based on users' characteristics, preferences, and health conditions. The study emphasized that AI-powered recommender systems help improve user engagement and encourage individuals to maintain healthy lifestyles by delivering recommendations that are more relevant to their needs. This study is relevant to the present study because it shows that artificial intelligence can provide personalized exercise recommendations based on users' fitness information, goals, and activity patterns,

allowing users to receive exercise guidance that is more suitable for their individual fitness needs.

Yin et al. (2022) examined the impact of gamification elements on user satisfaction in health and fitness applications using the Kano model. The study found that incorporating gamification into fitness applications significantly improves user satisfaction by making the application more engaging and enjoyable. The researchers also highlighted that users are more likely to continue using fitness applications when they include motivational and interactive features that encourage participation in physical activities. The results of the study serve as a basis for ASCENOVA by demonstrating that gamification features can improve user engagement and motivation, which are essential components of the proposed system's fitness tracking experience.

Gabarron et al. (2024) evaluated the comprehensiveness, accuracy, and readability of exercise recommendations generated by an artificial intelligence-based chatbot. The researchers found that the chatbot was capable of providing exercise recommendations that were generally accurate and easy to understand, although some recommendations lacked complete safety guidance and additional exercise details. The study emphasized that AI chatbots have the potential to support users in accessing exercise information and making informed decisions about their physical activities. The study

provides valuable insights to support the development of ASCENOVA by demonstrating that an artificial intelligence chatbot can assist users by providing interactive exercise guidance and personalized fitness recommendations.

Wang et al. (2024) conducted a systematic review and meta-analysis on mHealth application-based interventions for increasing physical activity and improving physical fitness in children and adolescents. The researchers found that, among children and adolescents, mobile health applications help increase physical activity by allowing users to monitor their progress, receive feedback, and stay engaged with their fitness goals. The study also concluded that mHealth applications can effectively encourage regular exercise and improve overall physical fitness among children and adolescents when combined with appropriate interventions. Although the study focused on children and adolescents, its results serve as a basis for ASCENOVA by emphasizing the importance of integrating fitness tracking, progress monitoring, and personalized exercise recommendations into a mobile-based fitness application that encourages users to maintain regular physical activity.

Local Literature

Batiquin (2025) examined the impact of mobile fitness applications on exercise adherence among Filipino youth. The study investigated how mobile fitness applications influence physical

activity, motivation, and long-term engagement among Filipino senior high school students. The findings showed that participants who used mobile fitness applications demonstrated improved exercise adherence, increased physical activity levels, and higher motivation to continue exercising. The study also identified gamification features, social interaction, and progress tracking as important factors that encouraged students to maintain their fitness routines. The study provides valuable insights for the development of ASCENOVA by showing that mobile fitness applications can improve users' motivation and encourage consistent participation in physical activities through personalized fitness tracking and gamification features.

Semsem and Martin (2022) designed and evaluated a mobile application for physical fitness testing that serves as an alternative tool for measuring the physical fitness levels of senior high school students. The researchers developed the application based on the Philippine Physical Fitness Test standards and evaluated its functionality with students, physical education teachers, and ICT experts. The study found that the mobile application was rated excellent in terms of functionality, interface, and usability, making it a practical tool for recording and computing students' physical fitness levels. This study contributes to the present research by emphasizing that mobile applications can improve fitness monitoring and provide

users with a more accessible and efficient way of tracking their physical fitness progress.

Obra et al. (2023) evaluated the effectiveness of using a mobile exercise application to improve exercise adherence and aerobic endurance among football players. The researchers found that participants demonstrated high adherence to the exercise program and showed improvements in aerobic endurance after completing the intervention. The study also highlighted that mobile exercise applications provide users with accessible workout guidance and encourage consistent participation in physical activities. The findings of the study support the development of ASCENOVA by demonstrating that mobile fitness applications can increase users' motivation and promote consistent exercise through guided workout programs and progress monitoring.

Tolentino et al. (2022) developed a mobile application for movement competency training using a user-centered design approach in a Philippine state university. The researchers emphasized that involving users throughout the development process helped create a mobile application that effectively supported physical education activities and movement competency training. The study highlighted that mobile technology can enhance learning experiences by providing accessible and interactive tools that encourage students to participate in physical activities. This study is

relevant to the present study because it demonstrates that user-centered mobile applications can effectively support physical activity while providing users with an engaging and accessible digital fitness experience.

Foreign Studies

Balpande et al. (2023) developed an AI-Based Gym Trainer and Diet Recommendation System that provides personalized exercise and diet recommendations using artificial intelligence. The system generates workout and diet suggestions based on users' physical activity and health-related information through a convolutional neural network. The researchers emphasized that personalized recommendations help users receive exercise guidance that is more appropriate to their individual needs instead of relying on general fitness advice. The results of the study serve as a basis for ASCENOVA by demonstrating that artificial intelligence can be utilized to generate personalized exercise recommendations based on users' fitness information and goals, which is one of the primary features of the proposed system.

Abdulaziz et al. (2021) developed a personalized fitness recommendation application that recommends suitable physical activities using users' personal information, such as heart rate, speed, and body measurements. The researchers applied the proposed recommendation system to the FitRec dataset and found

that it was capable of generating appropriate exercise recommendations for different groups of users based on their characteristics. The study highlighted that personalized recommendation systems can provide more relevant fitness guidance by considering users' individual information instead of applying the same exercise program to everyone. The study provides valuable insights for the development of ASCENOVA by showing that personalized exercise recommendations based on users' information can improve the relevance of suggested workouts and provide a more customized fitness experience.

Kothari et al. (2025) developed Wellness AI, an AI-driven platform that provides personalized diet and fitness recommendations by processing users' health information, lifestyle preferences, and fitness goals. The system integrates real-time health metrics from wearable devices and utilizes advanced large language models to generate personalized health insights and recommendations. The researchers highlighted that the platform enables users to receive more accurate and individualized guidance by analyzing multiple health-related factors. The results of the study serve as a basis for the development of ASCENOVA by demonstrating that artificial intelligence can be integrated into a fitness platform to deliver personalized exercise and nutrition

recommendations based on users' health information and lifestyle preferences.

Veerajeevanandam et al. (2026) proposed an AI-powered framework for generating personalized fitness and diet recommendations using users' clinical biomarkers and health information. The system utilizes the Gemini API together with Retrieval-Augmented Generation (RAG) to analyze health data, assess users' risk profiles, and generate personalized fitness and dietary recommendations. The researchers reported that the proposed framework improved personalization, usability, and user satisfaction by continuously refining recommendations based on user feedback. The study provides valuable insights for the development of ASCENOVA by showing that artificial intelligence can generate adaptive fitness and nutrition recommendations using users' health information while continuously improving recommendations through user feedback.

Local Studies

Ompoc and Aguinaldo (2025) evaluated the effectiveness of AI-driven applications in enhancing the physical fitness of tertiary physical education students in the Philippines. The researchers conducted an eight-week intervention that compared students who used AI-driven applications with those who followed traditional physical education activities. The findings revealed that students who

used AI-assisted applications demonstrated greater improvements in cardiovascular endurance, upper body strength, core strength, and lower body strength than those in the traditional group. The study also concluded that AI-driven applications can effectively support physical education by promoting better physical fitness and encouraging active participation among students. The results of the study serve as a basis for the development of ASCENOVA by demonstrating that artificial intelligence can improve users' fitness performance through personalized exercise guidance and digital fitness technologies.

Guadamor (2026) developed, implemented, and evaluated ChatFitPH, a localized AI-powered chatbot designed to improve the health and fitness literacy of senior high school learners. Using a mixed-methods research design, the study found significant improvements in learners' health literacy, fitness literacy, and engagement after one semester of chatbot use. The study also identified the importance of localization, gamification, accessibility, and personalized guidance in increasing user engagement and supporting healthier behaviors among students. The findings of the study support the development of ASCENOVA by demonstrating that AI-powered chatbots can effectively deliver personalized fitness guidance while improving user engagement through localized content and interactive features.

Cabauatan and Lopez (2021) compared the effectiveness of mobile fitness applications and gym exercises in improving the health-related fitness performance of first-year college students. Using a quasi-experimental research design, the researchers found that students who used mobile fitness applications achieved improvements in physical fitness that were comparable to those who performed traditional gym exercises. The study concluded that mobile fitness applications provide an effective alternative for promoting regular physical activity while encouraging users to develop healthier exercise habits. This study contributes to the present research by showing that mobile fitness applications can effectively encourage physical activity and improve users' fitness performance through digital exercise guidance and continuous progress monitoring.

Martin (2026) evaluated the effectiveness of the MyFitnessPal application in improving exercise adherence and aerobic endurance among senior high school students. The researchers conducted a six-week intervention and found that participants demonstrated high adherence to the exercise program and significant improvements in aerobic endurance after using the mobile application. The study concluded that fitness applications can serve as effective tools for promoting physical activity and supporting healthier lifestyles among students. This study is relevant to the

present study because it demonstrates that mobile fitness applications can improve exercise adherence while encouraging users to consistently participate in physical activities through digital fitness support.

Synthesis

The reviewed literature and studies consistently emphasize the importance of integrating artificial intelligence, mobile technology, and gamification in developing effective fitness applications. Both foreign and local literature highlighted that artificial intelligence enables personalized exercise recommendations based on users' fitness information, goals, and preferences, while gamification features such as points, achievements, challenges, and progress tracking improve user motivation and engagement. Likewise, the reviewed literature also showed that mobile fitness applications promote exercise adherence, support fitness monitoring, and encourage users to maintain healthier lifestyles through continuous participation in physical activities. On the other hand, the reviewed foreign and local studies demonstrated that intelligent fitness systems can provide personalized exercise recommendations, fitness monitoring, and AI-assisted coaching by utilizing users' health information and activity data. These findings indicate that combining intelligent technologies with interactive features can significantly enhance users' overall fitness experience.

Despite the positive findings presented in the reviewed literature and studies, most existing systems primarily focus on specific functionalities such as exercise recommendation, nutrition planning, fitness tracking, or AI chatbot assistance rather than integrating these features into a single fitness management system. The reviewed studies also show the potential of digital platforms to support fitness monitoring, user engagement, and personalized guidance. However, there remains a need for an integrated system that combines artificial intelligence, personalized exercise recommendations, fitness tracking, and gamification while providing separate interfaces for end-users and system administrators. Therefore, the researchers propose the development of ASCENOVA: An Intelligent Gamified Fitness Tracking and Personalized Exercise Recommendation System, consisting of a mobile application for end-users and a web application that provides the end-user features available on the mobile application, together with an admin dashboard for authorized administrators. The system aims to provide personalized fitness guidance and engagement features while allowing administrators to manage user accounts and achievements and to monitor user activity, system usage, and AI usage.

Conceptual Framework

The study adopts the Input-Process-Output (IPO) Model to illustrate the system development flow of ASCENOVA: An Intelligent Gamified Fitness Tracking and Personalized Exercise Recommendation System. The framework presents how the identified user, technical, hardware, and people requirements are transformed through an Agile development process into a mobile application and web application that integrate personalized exercise recommendations, fitness tracking, gamification, and administrative functions. It also incorporates an evaluation stage to assess the developed system based on the selected ISO/IEC 9126 software quality characteristics and the effectiveness of its major features.

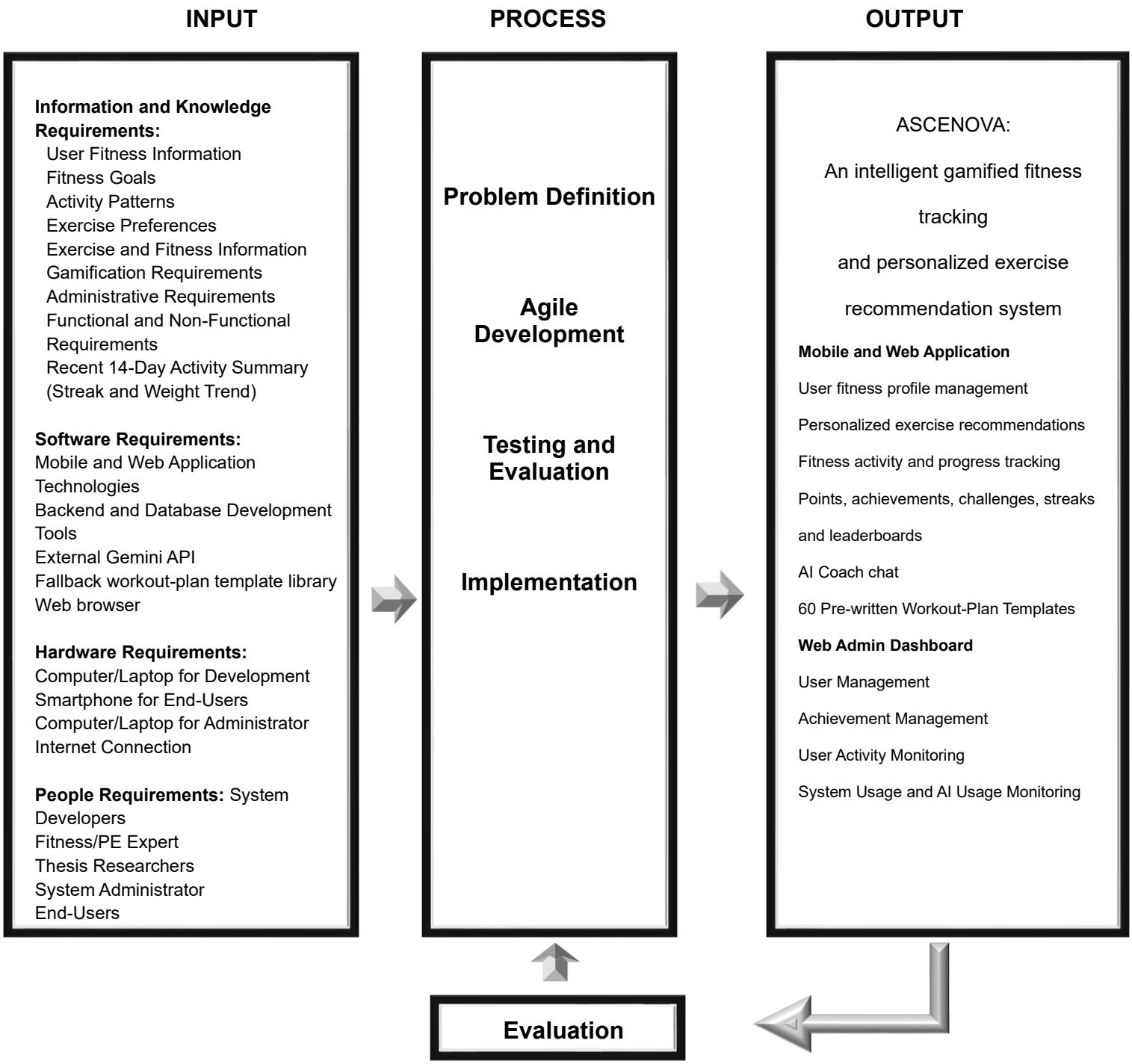


Figure 2.1 Conceptual Framework of the Study

Conceptual Framework of the Study

Figure 2.1 presents the conceptual framework of ASCENOVA using the Input-Process-Output (IPO) Model, which illustrates how system requirements are transformed through development into the proposed system and evaluated based on the study's objectives.

The Input component includes users' fitness information, fitness goals, activity patterns, exercise preferences, and the functional and non-functional requirements of ASCENOVA. It also covers the necessary software, hardware, development tools, Gemini API, Fallback Library of 60 Pre-written Workout-Plan Templates, and personnel involved in developing and evaluating the system. These inputs establish the requirements and resources needed for system development and serve as the basis for generating personalized exercise recommendations.

The Process component follows the Agile development approach, consisting of Problem Definition, Agile Development, Testing and Evaluation, and Implementation. Through iterative sprints, the researchers design, develop, test, and improve the system's features based on identified requirements and evaluation feedback before implementation.

The Output component is the ASCENOVA mobile application and web application, which provide user fitness profile management,

personalized exercise recommendations, fitness activity and progress tracking, and gamification features such as points, achievements, challenges, streaks, and leaderboards. The system also includes an AI Coach chat feature powered by the external Gemini API and a fallback library of 60 pre-written workout-plan templates for use when the AI service fails to generate a workout plan. The web application includes an admin dashboard for managing user accounts and achievements and monitoring user activity, system usage, and AI usage.

The Evaluation component assesses the system using six selected ISO/IEC 9126 software quality characteristics: functionality, efficiency, usability, reliability, maintainability, and portability. Evaluation questionnaires, weighted mean, and feedback from a qualified fitness expert are used to assess system quality and the appropriateness of exercise recommendations. The results guide necessary improvements to the system.

Overall, the framework illustrates how ASCENOVA transforms identified requirements through Agile development into an integrated fitness management system, with evaluation supporting the assessment and continuous improvement of its features.

OPERATIONAL DEFINITION OF TERMS

Artificial Intelligence (AI)	Refers to the technology used by ASCENOVA to analyze users' fitness information and generate personalized exercise recommendations.
ASCENOVA	Refers to the intelligent gamified fitness tracking and personalized exercise recommendation system developed in this study.
Exercise Recommendation	Refers to the exercise suggestions generated by ASCENOVA based on users' fitness information, goals, and activity patterns.
Fitness Tracking	Refers to the process of recording and monitoring users' workout activities and fitness progress through the system.
Gamification	Refers to the use of game-like features such as points,

	achievements, challenges, streaks, leaderboards, and progress tracking to encourage users to stay motivated and engaged.
ISO/IEC 9126	Refers to the international software product quality model used as the basis for evaluating six selected quality characteristics of the ASCENOVA system: Functionality, Efficiency, Usability, Reliability, Maintainability, and Portability.
Personalized Exercise Recommendation	Refers to exercise recommendations that are tailored according to the user's fitness information, goals, preferences, and activity patterns.
Fitness Information	Refers to the fitness-related information provided by users

	and used by ASCENOVA for generating exercise recommendations.
Fitness Goals	Refers to the specific fitness objectives identified by users and considered by ASCENOVA when generating exercise recommendations.
Activity Patterns	Refers to the user's recorded exercise activities and patterns of physical activity considered by the system for personalized recommendations.
Fitness Progress	Refers to the changes and improvements in the user's recorded fitness activities and performance over time.
Portability	Refers to the ability of the ASCENOVA system to adapt to different or changing usage, operational, or environmental

	needs, as applicable to its intended use.
Progress Tracking	Refers to the feature of ASCENOVA that allows users to monitor their exercise activities and progress toward their fitness goals.
User Satisfaction	Refers to the level of satisfaction of users with the features, recommendations, usability, and overall experience provided by ASCENOVA.
Functionality	Refers to the ability of ASCENOVA to provide the functions needed for personalized exercise recommendations, fitness tracking, gamification, and administrative management, including the protection of user accounts and personal fitness information against

	unauthorized access and misuse.
Efficiency	Refers to the ability of ASCENOVA to perform its functions and respond to user actions within an appropriate amount of time and with reasonable use of system resources.
Usability	Refers to the ability of ASCENOVA to allow users to interact with the system effectively and conveniently through understandable navigation, clear interface elements, and accessible functions.
Reliability	Refers to the ability of ASCENOVA to perform its intended functions consistently

	and correctly. including the use of the fallback workout-plan library when the external AI service fails to generate a workout plan.
Maintainability	Refers to the ability of ASCENOVA to be analyzed, corrected, and modified without unnecessarily affecting other system components.
Mobile Application	Refers to the ASCENOVA application used by end-users to access personalized exercise recommendations, record fitness activities, monitor progress, and use gamification features.
Web Application	Refers to the web-based component of ASCENOVA that provides the end-user features available on the mobile application, together with an

	admin dashboard that authorized administrators use to manage user accounts and achievements and to monitor user activity, system usage, and AI usage.
Administrator	Refers to the authorized individual responsible for managing and monitoring relevant information through the admin dashboard of the ASCENOVA web application.
Fitness Expert	Refers to the qualified fitness or physical education professional who assists in reviewing and validating the appropriateness of the exercise recommendations provided by ASCENOVA.

AI Service/API	Refers to the external artificial intelligence service accessed by ASCENOVA to process user fitness information and generate personalized exercise recommendations.

Chapter 3

RESEARCH METHODOLOGY

This chapter presents the methods and procedures used in the design, development, and evaluation of ASCENOVA: An Intelligent Gamified Fitness Tracking and Personalized Exercise Recommendation System. It discusses the project design, project development process, testing and evaluation, statistical treatment of data, and implementation of the proposed system. It also describes the methods used to evaluate the developed system based on the identified software quality characteristics and system features.

Research Method

This study will utilize a developmental research approach. This research method is appropriate because the main purpose of the study is to design, develop, and evaluate ASCENOVA: An Intelligent Gamified Fitness Tracking and Personalized Exercise Recommendation System. The study focuses on developing a mobile application for end-users and a web application that provides the end-user features available on the mobile application, together with an admin dashboard for authorized administrators.

The researchers will adopt an Agile development approach to support continuous development and improvement of the system through short development cycles called sprints. Each sprint involves planning, requirements and analysis, design and development,

testing, evaluation and feedback, and improvement. The results and feedback obtained from each sprint will be used to improve the succeeding development cycle.

Design of the Study

The design of the study follows an Agile Software Development Life Cycle (SDLC), which supports incremental and continuous development through a series of sprints. Each sprint focuses on selected system requirements and features that undergo planning, requirements and analysis, design and development, testing, evaluation and feedback, and improvement.

The Agile approach allows the researchers to develop the ASCENOVA mobile application and web application incrementally. Feedback and issues identified during testing and evaluation will be considered in making necessary revisions and improvements before proceeding to the succeeding sprint. This process allows the system features to be continuously developed, tested, and improved throughout the development process.

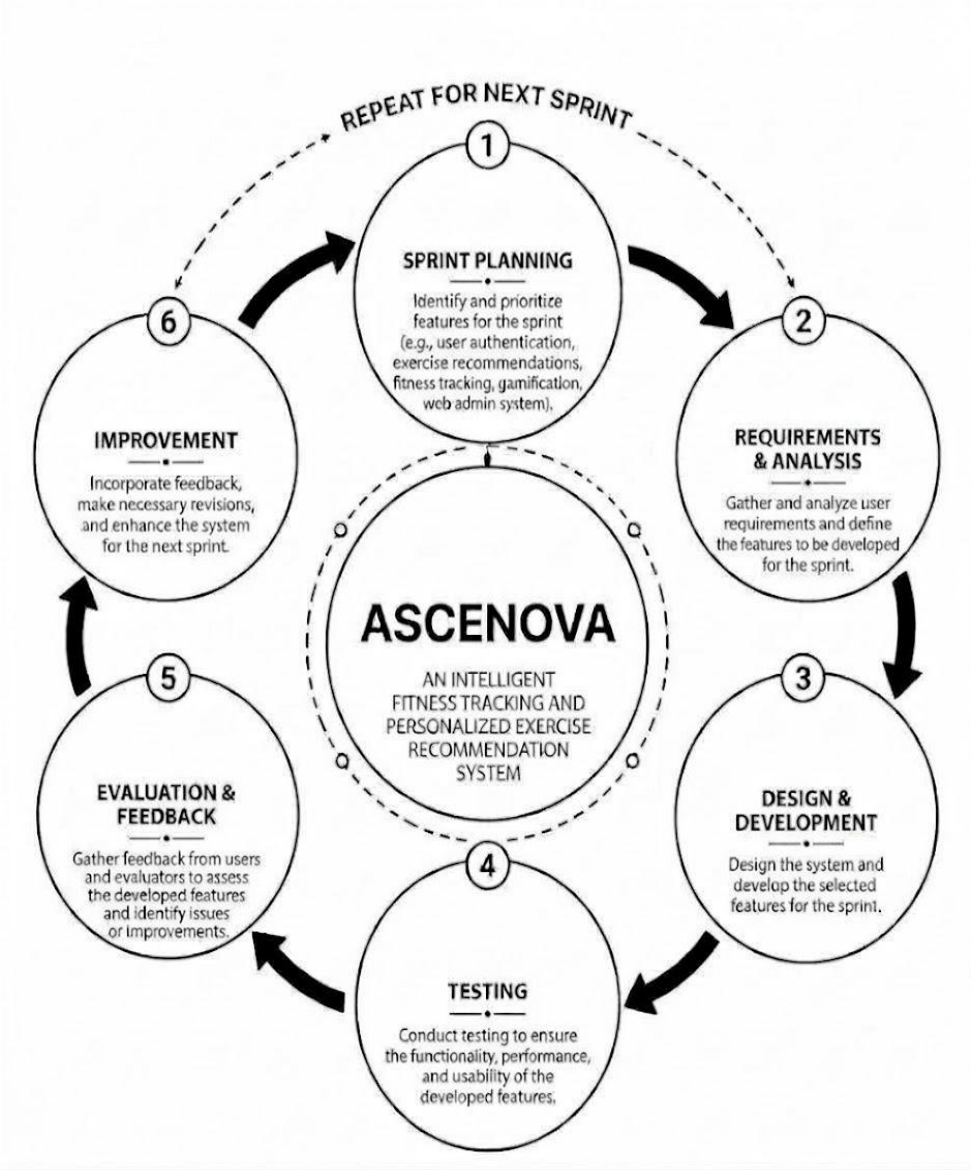


Figure 3.1 Agile Sprint Cycle Diagram

Sprint Planning

The researchers will identify and prioritize the system features and requirements to be developed in each sprint. The scope and objectives of the sprint will be established based on the identified requirements of ASCENOVA.

Requirements and Analysis

The researchers will gather and analyze the requirements needed for the selected features of ASCENOVA. The requirements will include the needs of the mobile application, web application, personalized exercise recommendations, the AI Coach chat, the fallback template library, fitness tracking, progress monitoring, and gamification features.

Design and Development

The researchers will design and develop the selected features for each sprint using the identified technologies and development tools. This phase includes the development of the ASCENOVA mobile application and web application and the integration of the required system components, the external artificial intelligence service/API, and the fallback template library.

Testing

Each developed feature will undergo testing to identify errors, bugs, and issues that may affect its functionality, including security, as well as its efficiency, usability, and reliability. The identified issues will be documented for correction and improvement.

Evaluation and Feedback

The developed features will be evaluated based on the applicable system requirements and evaluation criteria. Feedback obtained from testing, evaluation, and the review of the personalized exercise

recommendations will be considered in identifying necessary improvements to the system.

Improvement

The researchers will use the results and feedback from the previous stages to make necessary revisions and enhancements to the developed features. The improved features will be incorporated into the succeeding sprint, allowing ASCENOVA to undergo continuous development and improvement until the required system features are completed and evaluated.

Research Design

The research design focuses on the evaluation of the developed ASCENOVA system. The evaluation will utilize a quantitative approach through an evaluation questionnaire based on selected ISO/IEC 9126 software quality characteristics (ISO/IEC, 2001). Although ISO/IEC 9126-1:2001 has been superseded by ISO/IEC 25010, it was adopted in this study per institutional requirement. The collected responses will be analyzed using descriptive statistical methods, particularly the weighted mean, to determine the overall assessment of the developed system.

The study will also involve qualitative feedback from a qualified fitness expert to assess the appropriateness and relevance of the personalized exercise recommendations generated by

ASCENOVA. The feedback will be used to identify possible improvements in the system's recommendations and features.

PROJECT DESIGN

The project design for ASCENOVA focuses on the development of an intelligent and gamified fitness system consisting of a mobile application for end-users and a web application that provides the end-user features available on the mobile application, together with an admin dashboard for authorized administrators. The system is designed to provide personalized exercise recommendations, fitness activity tracking, progress monitoring, gamification features, and administrative functions.

The design is based on the identified functional and non-functional requirements of the system. It integrates an external artificial intelligence service/API to generate personalized exercise recommendations based on users' fitness information, goals, preferences, and activity patterns. The same service powers the AI Coach chat feature. As a reliability feature, the system includes a fallback library of 60 pre-written workout plan templates, matched to the user's goal, fitness level, equipment, and preferred training days, which is used when the external service fails to generate a plan. The project design serves as the basis for the development, testing, and evaluation of the ASCENOVA system.

Technology Stack

ASCENOVA will be developed using a technology stack consisting of TypeScript, Express.js, Node.js, PostgreSQL hosted on Supabase, Prisma, React with Vite, React Native with Expo, and the Gemini API. The technology stack is composed of the following components:

- Express.js (Backend Framework) – used to build the server-side application programming interfaces (APIs) that handle requests between the frontend and the database.
- TypeScript (Programming Language) – used to write the backend server of ASCENOVA.
- React with Vite and React Native with Expo (Frontend) – used to build the user interfaces of the ASCENOVA web application, which includes the admin dashboard, and the mobile application, respectively.
- Node.js (Runtime Environment) – used to run the backend server and handle the application's server-side logic.
- PostgreSQL and Supabase (Database) – used to provide the system's relational database, with PostgreSQL hosted on Supabase.
- Prisma (Object-Relational Mapping) – used by the backend server to access the PostgreSQL database.

- Gemini API (External Artificial Intelligence Service) – used to process users' fitness information, goals, preferences, and activity patterns in order to generate personalized exercise recommendations and to power the AI Coach chat.

ASCENOVA will integrate the Gemini API as its external artificial intelligence service to process users' fitness information and generate personalized exercise recommendations based on the identified functional and non-functional requirements of the system. If the Gemini API fails to generate a plan, a fallback library of pre-written workout plan templates is used instead. For security, the system uses custom JSON Web Token (JWT) access tokens with refresh tokens, role-based access control for the USER and ADMIN roles, and soft deletion of user accounts.

Tools, Methods, and Algorithms

The development of ASCENOVA utilizes a combination of frontend, backend, database, and artificial intelligence technologies organized within a modular application architecture. The mobile application is developed using React Native with Expo, while React with Vite is used for the web application, including the administrative dashboard. TypeScript is used for backend development, with Node.js serving as the runtime environment and Express.js handling server-side application programming interfaces (APIs). Prisma is

used to facilitate database operations with PostgreSQL hosted on Supabase.

The system integrates the external Google Gemini API to generate personalized exercise recommendations based on users' fitness information, goals, exercise preferences, and activity patterns. When a user requests a new workout plan, the system also considers a summary of the user's activity over the previous 14 days, including streak and weight trend, to support more relevant recommendations. The same external AI service powers the AI Coach chat feature. To maintain workout plan availability when the external AI service fails to generate a plan, ASCENOVA incorporates a fallback library of 60 pre-written workout-plan templates matched to the user's fitness goal, fitness level, available equipment, and preferred training days.

For authentication and access control, ASCENOVA uses custom JSON Web Token (JWT) access tokens with refresh tokens and role-based access control for USER and ADMIN roles. These mechanisms support authenticated access to system features according to the user's assigned role. The system also implements soft deletion of user accounts. For system development, the researchers adopt an Agile approach that involves planning,

requirements analysis, design and development, testing, evaluation and feedback, and improvement through successive sprints. These methods support the incremental development, testing, and refinement of ASCENOVA's mobile application, web application, personalized exercise recommendations, fitness tracking, gamification, and administrative functions.

Project Development

The project development follows an Agile development approach based on the requirements identified for ASCENOVA. The development process includes: **(1) problem definition, (2) Agile sprint development, (3) testing and evaluation, and (4) system implementation.**

These stages guide the researchers in transforming the identified problems and system requirements into a functional fitness management system. The development process covers the creation of the ASCENOVA mobile application and web application, integration of the external artificial intelligence service/API and the fallback template library, implementation of fitness tracking and gamification features, testing of the system functions, and necessary improvements based on feedback and evaluation results from each sprint.

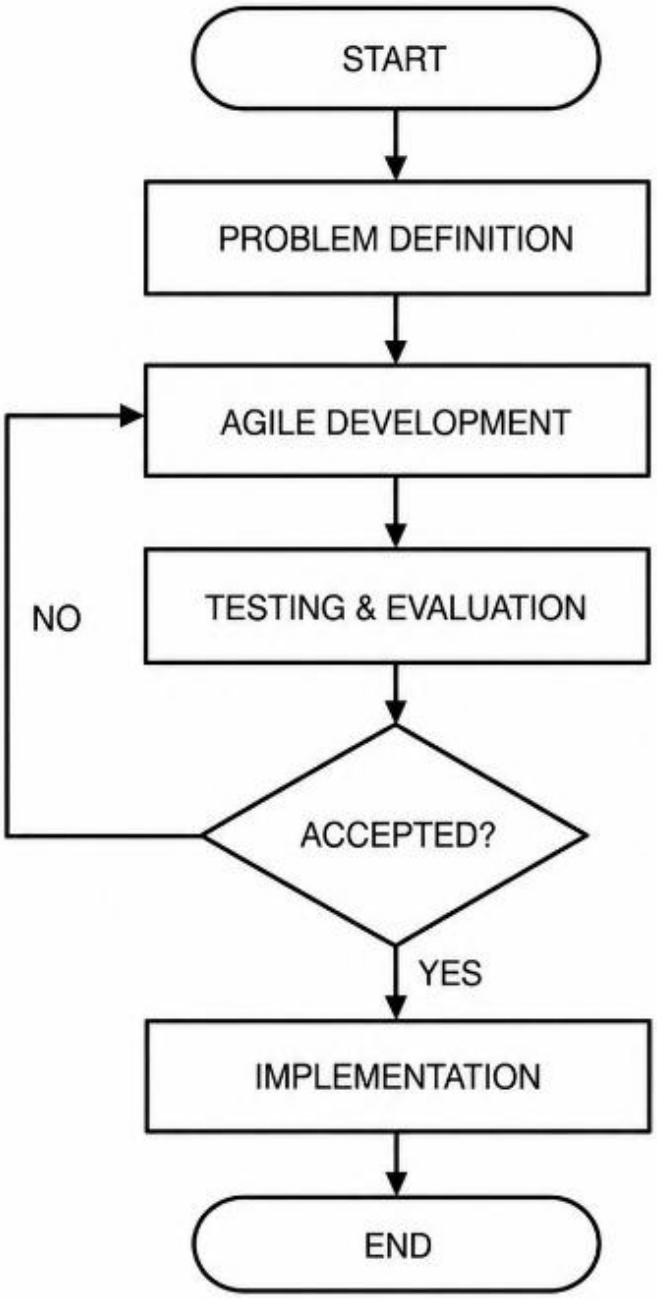


Figure 3.2 Project Development Flow Diagram

Figure 3.2 presents the project development flow of the proposed ASCENOVA system. The process begins with the Problem Definition stage, where the researchers identify the needs, issues, and requirements related to personalized exercise recommendations, fitness tracking, gamification, and administrative functions. This is followed by Agile Development, where the identified requirements and selected system features are developed incrementally through a series of sprints. Each sprint involves planning, requirements and analysis, design and development, testing, evaluation and feedback, and improvement.

After the developed features undergo testing and evaluation, the researchers determine whether the required system features have been achieved. If revisions are required, the process returns to Agile Development for further improvement in the succeeding sprint. Once the system is accepted, it proceeds to the Implementation stage, where the completed mobile application and web application are prepared for deployment and use.

Problem Defintion

The Problem Definition stage identifies the issues and needs that support the development of ASCENOVA. Existing fitness applications commonly provide exercise libraries that offer different exercises for specific body parts, such as the legs, back, chest, and arms. However, these applications may provide general exercise

options without considering the individual fitness information, goals, preferences, and activity patterns of each user.

This limitation may make it difficult for users to determine which exercises are most suitable for their individual fitness needs. The researchers identified the need for a fitness system that goes beyond providing a general list of exercises by offering personalized exercise recommendations based on the user's fitness information, goals, preferences, and activity patterns.

ASCENOVA addresses this need by combining personalized exercise recommendations with fitness activity tracking and gamification features such as points, achievements, challenges, streaks, leaderboards, and progress tracking. The system also includes a web application with an admin dashboard that allows authorized administrators to manage user accounts and achievements and to monitor user activity, system usage, and AI usage.

Based on these identified needs, the researchers proposed ASCENOVA: An Intelligent Gamified Fitness Tracking and Personalized Exercise Recommendation System to provide users with a more personalized and engaging approach to fitness management.

Testing and Evaluation

In this phase, the ASCENOVA mobile application and web application will be tested to identify possible bugs, errors, and issues that may affect the proper functioning of the system. Different types of testing will be conducted to evaluate the system's functionality, including security, as well as its efficiency, usability, and reliability. The testing will also assess the personalized exercise recommendations, fitness tracking, progress monitoring, gamification features, and administrative functions. The fallback template library will also be tested as part of reliability to verify that workout plans are still provided when the external artificial intelligence service fails.

The personalized exercise recommendations will also be reviewed with the assistance of a qualified fitness expert to assess their appropriateness and relevance based on the users' fitness information, goals, preferences, and activity patterns.

Respondents of the Study

The respondents of the study will consist of thirty (30) selected participants, comprising three (3) IT professionals and twenty-seven (27) non-IT respondents. The non-IT respondents will include college students who engage in fitness activities, gym-goers, fitness enthusiasts, and fitness coaches, with one qualified fitness expert included among the twenty-seven (27) non-IT respondents. The IT

professionals will evaluate the technical aspects and software quality of the developed ASCENOVA system, particularly its functionality, efficiency, usability, reliability, maintainability, and portability. The non-IT respondents will evaluate the system's usability and user-facing features, including personalized exercise recommendations, fitness activity and progress tracking, and gamification. The qualified fitness expert will also review the appropriateness and relevance of the personalized exercise recommendations generated by the system.

Sampling Technique

The study will use purposive sampling, a non-probability sampling technique, to select thirty (30) respondents who meet the criteria for evaluating the developed ASCENOVA system. The respondents will consist of three (3) IT professionals and twenty-seven (27) non-IT respondents. The IT professionals will be selected based on their relevant technical knowledge and expertise in evaluating software quality. The non-IT respondents will include college students who engage in fitness activities, gym-goers, fitness enthusiasts, and fitness coaches who can assess the system's usability and fitness-related features. One qualified fitness expert will be included among the twenty-seven (27) non-IT respondents to review the appropriateness and relevance of the personalized exercise recommendations generated by ASCENOVA. This sampling

technique will ensure that the selected participants can provide relevant feedback based on their respective knowledge, experience, and areas of expertise.

Data Gathering Procedure

The data needed for the evaluation of ASCENOVA will be gathered through online evaluation questionnaires administered using Google Forms. A total of thirty (30) respondents will participate in the evaluation, consisting of three (3) IT professionals and twenty-seven (27) non-IT respondents, including college students who engage in fitness activities, gym-goers, fitness enthusiasts, and fitness coaches. One qualified fitness expert will be included among the non-IT respondents.

Before answering the questionnaires, the respondents will be provided access to the developed ASCENOVA mobile application and web application to explore and test the features relevant to their roles. They will then receive a Google Forms link containing the appropriate evaluation questionnaire, which they will complete based on their experience using the system.

The evaluation questionnaires will be organized according to the respondents' respective roles and areas of assessment. The IT Professional Evaluation will focus on the technical aspects and

selected software quality characteristics of ASCENOVA, including functionality, efficiency, usability, reliability, maintainability, and portability. The Non-IT Respondent Evaluation will focus on the user-facing features of ASCENOVA, including personalized exercise recommendations, fitness activity and progress tracking, gamification features, and overall usability.

The qualified fitness expert will additionally review the personalized exercise recommendations generated by ASCENOVA to assess their appropriateness and relevance to users' fitness information, goals, preferences, and activity patterns. The expert's feedback will be documented and used to identify possible improvements to the recommendations.

The quantitative evaluation responses will be compiled, organized, and analyzed using the weighted mean to determine the respondents' assessments of the system based on the selected ISO/IEC 9126 software quality characteristics and its major features. The fitness expert's review will be considered separately as additional feedback on the exercise recommendations. Before participating, all respondents will be informed of the study's purpose and asked to provide informed consent. Their responses will be treated confidentially and used solely for academic purposes. The

findings will serve as a basis for identifying areas for improvement and formulating recommendations for the further enhancement of ASCENOVA.

Ethical Considerations

The study will observe ethical standards throughout the conduct of the research. Respondents will be informed of the purpose of the study, and their participation in the evaluation of ASCENOVA will be entirely voluntary. Informed consent will be secured from all respondents prior to their participation, and they may withdraw at any point without penalty.

All information and personal data gathered through the online questionnaire will be treated with strict confidentiality and will be used solely for the purposes of this study. No personally identifiable information will be disclosed in the presentation of the results, and collected data will be securely stored and accessed only by the researchers.

The study will also comply with the Data Privacy Act of 2012, Republic Act No. 10173. Personal data will be collected only with the informed consent of the respondents, used only for the stated purposes of the study, and protected against unauthorized access, disclosure, or misuse.

Statistical Treatment of the Data

The collected evaluation data will be organized, summarized, and analyzed using descriptive statistical methods. The weighted mean will be used to determine the overall assessment of the respondents for each evaluation criterion and the major features of ASCENOVA.

Weighted Mean

The weighted mean will be used to determine the average rating of the respondents while considering the frequency of each response.

Weighted Mean = Σ(fx) / n

Where:

Σ = Summation

f = Frequency of responses

x = Rating value

n = Total number of responses

The weighted mean will be used to determine the overall rating of the selected ISO/IEC 9126 software quality characteristics and major features of ASCENOVA.

Likert Scale

The evaluation questionnaire will use a five-point Likert scale to assess respondents' perceptions of ASCENOVA's selected software quality characteristics and major system features. Each statement will be rated from 1 to 5, with higher scores indicating a more

favorable assessment of the system. The weighted mean will be used to summarize the responses for each evaluation criterion.

Table 3.1 Likert Scale

Range	Interpretation
4.21 - 5.00	Highly Satisfied
3.41 - 4.20	Satisfied
2.61 - 3.40	Neutral
1.81 - 2.60	Dissatisfied
1.00 - 1.80	Highly Dissatisfied

The respondents will rate each criterion using a scale from 1 to 5, with a higher rating indicating a more favorable assessment of the system. The ratings will then be analyzed using the weighted mean to determine the overall evaluation of ASCENOVA.

ISO/IEC 9126 Evaluation Criteria

The evaluation will focus on six selected ISO/IEC 9126 software quality characteristics to assess the quality of the developed ASCENOVA system:

1. Functionality – assesses whether ASCENOVA provides the functions needed for fitness tracking, exercise recommendations, gamification, and administrative functions, and whether it protects user information and prevents unauthorized access.

2. Efficiency – assesses whether the system performs its functions and responds to user actions within an appropriate amount of time.
3. Usability – assesses whether users can understand, learn, and use ASCENOVA effectively and conveniently.
4. Reliability – assesses whether ASCENOVA performs its intended functions consistently and correctly.
5. Maintainability – assesses whether ASCENOVA can be analyzed, corrected, and modified without unnecessarily affecting other system components.
6. Portability – assesses whether ASCENOVA can be adapted to support its intended use across the mobile application and the web application.

These criteria are based on the operational definitions established in Chapter II.

Implementation

The implementation of ASCENOVA will focus on deploying the developed mobile application and web application with their key features fully functional. The mobile application will provide personalized exercise recommendations, fitness tracking, progress monitoring, and gamification features for end-users, while the web application will provide the end-user features available on the mobile application together with administrative functions, through its admin dashboard, for managing user accounts and achievements and for monitoring user activity, system usage, and AI usage. The proposed system will be implemented according to the identified requirements and limitations of the study.

References

Kothari, A., Khanna, U., Shah, N., & Shah, S. (2025). Wellness AI: AI-driven personalized diet and fitness recommendation. In 2025 IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECS). IEEE. https://doi.org/10.1109/SCEECS64059.2025.10940178

Veerajeevanandam, V., Santhosh Perumal, S., & Percy Paulin, J. (2026). AI-powered framework for personalized fitness and diet recommendations using clinical biomarkers. In 2026 International Conference on Innovative Trends in Information Technology (ICITIIT). IEEE. https://doi.org/10.1109/ICITIIT68860.2026.11499497

Tolentino, J. C. G., Miranda, J. P. P., Punzalan, R. B., Manalang, J. C., Hermogenes, L. K. S., & Mallari, J. T. (2022). Towards the development of a mobile application in movement competency training grounded on the user-centered design model: The case of a state university in the Philippines. International Journal of Interactive Mobile Technologies (iJIM), 16(3), 92–103. https://doi.org/10.3991/ijim.v16i03.26447

Semsem, K., & Martin, J. T. (2022). Development of a mobile application for physical fitness testing. International Journal of Human Movement and Sports Sciences, 10(6), 1126–1133. https://doi.org/10.13189/saj.2022.100603

Martin, B. J. M. (2026). Effectiveness of MyFitnessPal among the senior high school students. *The Physical Educator, 83*(2), 189–209. https://doi.org/10.18666/TPE-2026-V83-I2-12823

Obra, C. D. N. N., Quidilla, A. C. S., Cajigal, R. J. B., Jerez, J. K. A., Ramos, S. M. P., & Malabed, R. J. C. (2023). Effectiveness of using exercise app in improving exercise adherence and aerobic endurance of football players. Philippine Journal of Physical Therapy, 2(2), 4–11. https://doi.org/10.46409/002.QFAG7056

Guadamor, T. J. B. (2026, March). *ChatFitPH: Building a conversational bot for Filipino learners' health and fitness needs* [Conference paper]. North Cluster Final Research Colloquium. https://www.researchgate.net/publication/402153850_ChatFitPH_Building_a_Conversational_Bot_for_Filipino_Learners'_Health_and_Fitness_Needs

Gabarron, E., et al. (2024). Comprehensiveness, accuracy, and readability of exercise recommendations provided by an AI-based chatbot: Mixed methods study. JMIR Medical Education, 10, e51308. https://mededu.jmir.org/2024/1/e51308/

Wang, J.-W., Zhu, Z., Zhang, S., et al. (2024). Effectiveness of mHealth app-based interventions for increasing physical activity and improving physical fitness in children and adolescents: Systematic

review and meta-analysis. JMIR mHealth and uHealth, 12, e51478. https://mhealth.jmir.org/2024/1/e51478

Cabauatan, K. A. F., & Lopez, G. A. D. (2021). Mobile fitness applications and gym exercises of first-year college students. https://www.globusedujournal.in/wp-content/uploads/2021/11/GE-JD21-King-Abram-F.-Cabauatan.pdf

Ompoc, H. M. S., & Aguinaldo, J. C. (2025). Evaluating the Effectiveness of AI-Driven Applications in Enhancing the Physical Fitness of Tertiary Physical Education Students. International Journal of Human Movement and Sports Sciences, 13(1), 74–82. https://doi.org/10.13189/saj.2025.130109

Balpande, M., Sharma, J., Nair, A., Khandelwal, M., & Dhanray, S. (2023). AI Based Gym Trainer and Diet Recommendation System. In 2023 IEEE 4th Annual Flagship India Council International Subsections Conference (INDISCON). IEEE. https://doi.org/10.1109/INDISCON58499.2023.10270066

Abdulaziz, M., Al-motairy, B., Al-ghamdi, M., & Al-qahtani, N. (2021). Building a Personalized Fitness Recommendation Application based on Sequential Information. International Journal of Advanced Computer Science and Applications, 12(1). https://doi.org/10.14569/IJACSA.2021.0120173

Batiquin, M. (2025). Impact of mobile fitness apps on exercise adherence among Filipino youth. Edelweiss Applied Science and Technology, 9(10), 1102–1129. https://doi.org/10.55214/2576-8484.v9i10.10603

Lopez-Barreiro, J., García-Soidán, J. L., Álvarez-Sabucedo, L., & Santos-Gago, J. M. (2024). Artificial intelligence-powered recommender systems for promoting healthy habits and active aging: A systematic review. Applied Sciences, 14(22), 10220. https://doi.org/10.3390/app142210220

World Health Organization. (2022). Global status report on physical activity 2022. World Health Organization. https://www.who.int/publications/i/item/9789240059153

Yin, S., Cai, X., Wang, Z., Zhang, Y., Luo, S., & Ma, J. (2022). *Impact of gamification elements on user satisfaction in health and fitness applications: A comprehensive approach based on the Kano model*. **Computers in Human Behavior, 128**, 107106. https://doi.org/10.1016/j.chb.2021.107106 .

ISO/IEC. (2001). *Software engineering: Product quality. Part 1: Quality model* (ISO/IEC 9126-1:2001). International Organization for Standardization.